

Dupire PINN / PDF validation

KDE vs NN density: IBP mean diagnostic

Ameir Shaa

LocalVolatility PINN pipeline

May 26, 2026

Setup: Dupire PINN pipeline

Networks:

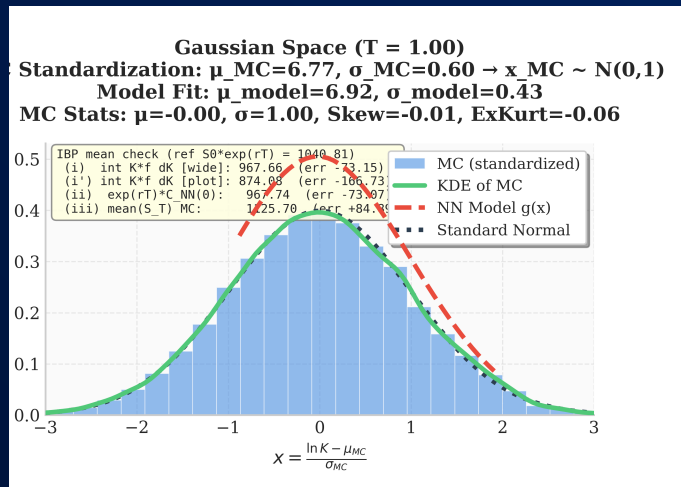
- NN_phi: normalized call price $\tilde{\varphi}(T, K)$
- NN_eta: local volatility squared $\tilde{\eta}(T, K)$

PDF analysis (three panels per maturity T):

1. Strike space $f(K)$ — MC histogram vs NN density
2. Log-strike $f(\ln K)$
3. Gaussian space — KDE of MC vs NN (standardized)

Training map: $\tilde{\varphi} = 1 - e^{-\text{NN}_\phi}$ (not raw NN_ϕ).

What Nicolas sees ($T = 1.0$, pretrained)



Gaussian panel: KDE of MC (blue) vs NN model (orange). Visible shape mismatch motivates the mean check.

Nicolas's concern (integration by parts)

Mean of the NN-implied risk-neutral density:

$$e^{rT} \int_0^\infty K \frac{\partial^2 C_{\text{NN}}}{\partial K^2} dK$$

Integration by parts (with $C \rightarrow 0$ as $K \rightarrow \infty$):

$$= e^{rT} C_{\text{NN}}(0, T) = \mathbb{E}[S_T]$$

Under no-arbitrage, all three should equal the forward $S_0 e^{rT}$.

Question

Where does the visible KDE–NN deviation come from if these means agree?

Three-way diagnostic test

Compute separately at each maturity T :

Label	Quantity
(i)	$e^{rT} \int_0^\infty K \frac{\partial^2 C_{\text{NN}}}{\partial K^2} dK$
(ii)	$e^{rT} C_{\text{NN}}(0, T)$
(iii)	$\mathbb{E}[S_T]$ from MC with $\sigma_{\text{NN}}(t, S)$
ref	$S_0 e^{rT}$

Implemented in `compute_mean_diagnostics`; yellow box on Panel 3. For (i) use **unclipped** $\partial^2 C / \partial K^2$ (no normalise-to-1).

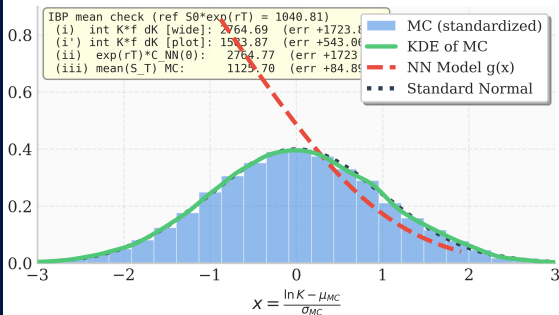
Finding 1: IBP holds in our code

At every T , (i) $\equiv$ (ii) to machine precision ($\sim 6 \times 10^{-5}$ relative).

- Autodiff and numerical integration are consistent
- The breakdown is **not** a wrong $\partial^2 C / \partial K^2$ formula

Pre-correction: legacy $\tilde{\varphi}$ bug ($T = 1.0$)

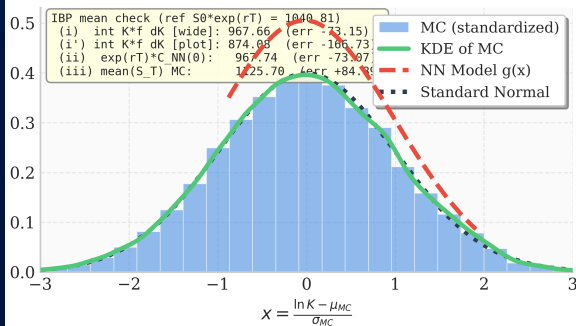
Gaussian Space ($T = 1.00$)
Standardization: $\mu_{MC}=6.77, \sigma_{MC}=0.60 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.73, \sigma_{model}=0.44$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.06$



Analysis used raw NN_ϕ as $\tilde{\varphi}$ instead of $1 - e^{-NN_\phi}$. Inflates (i) and (ii) $\approx 2.7 \times$ ref.

Pretrained: corrected $\tilde{\varphi}$ ($T = 1.0$ row)

Gaussian Space ($T = 1.00$)
Standardization: $\mu_{MC}=6.77$, $\sigma_{MC}=0.60 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92$, $\sigma_{model}=0.43$
MC Stats: $\mu=-0.00$, $\sigma=1.00$, Skew=-0.01, ExKurt=-0.06



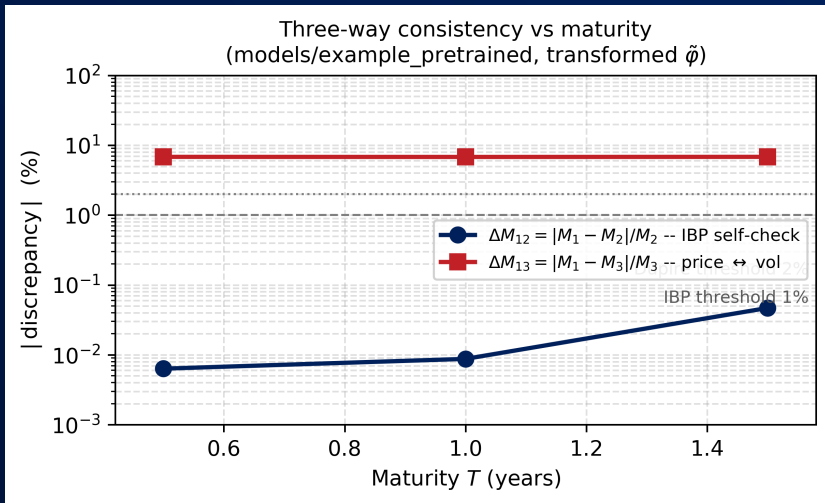
Panel 3: yellow box lists (i), (i'), (ii), (iii) vs ref. Same MC paths as pre-correction slide.

Three-way check (synthetic baseline, models/example_pretrained)

T	M_{ref}	M_1	%err	M_2	%err	M_3	%err	$\Delta M_{12}\%$	$\Delta M_{13}\%$
0.5	1020.20	948.87	-6.99	948.93	-6.99	1018.25	-0.19	0.006	6.814
1.0	1040.81	967.66	-7.03	967.74	-7.02	1038.19	-0.25	0.009	6.794
1.5	1061.84	982.49	-7.47	982.95	-7.43	1054.36	-0.70	0.047	6.816

- $M_1 \equiv M_2$ to $< 0.05\%$ everywhere $\Rightarrow$ IBP identity exact.
- M_3 within 0.7% of forward $\Rightarrow$ NN_η MC well-calibrated.
- $M_1, M_2 \approx 7\%$ below forward $\Rightarrow$ NN_φ undershoots near $K = 0$.

ΔM_{12} and ΔM_{13} as a function of maturity



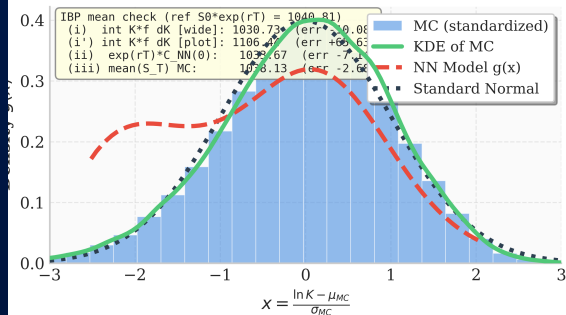
ΔM_{12} is at numerical-noise level (IBP exact); $\Delta M_{13} \approx 7\%$ across all maturities $\Rightarrow$ price $\leftrightarrow$ vol gap concentrated in NN_{ϕ} at small K .

Solution: $K = 0$ boundary penalty + wider strikes

- Penalty $|C_{\text{NN}}(0, T) - S_0|$ in `loss_phi_cal` ($\lambda_{K0} = 10$)
- Training strikes $K_{\min} = 100$ (was 500)
- Retrain $\rightarrow$ `models/runs/k0_bc_retrain`

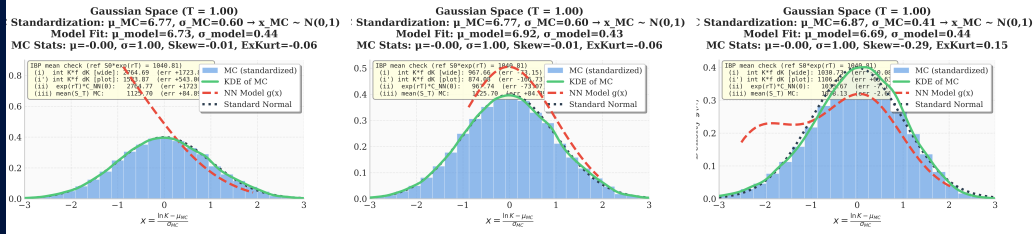
Retrained: K_0 BC ($T = 1.0$)

Gaussian Space ($T = 1.00$)
Standardization: $\mu_{MC}=6.87, \sigma_{MC}=0.41 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.69, \sigma_{model}=0.44$
MC Stats: $\mu=-0.00, \sigma=1.00$, Skew=-0.29, ExKurt=0.15



$T = 1.0$: (i)=1030.7, (ii)=1033.7, (iii)=1038.1, ref=1040.8 (within $\sim 1\%$).

Before / after: Gaussian panel at $T = 1.0$



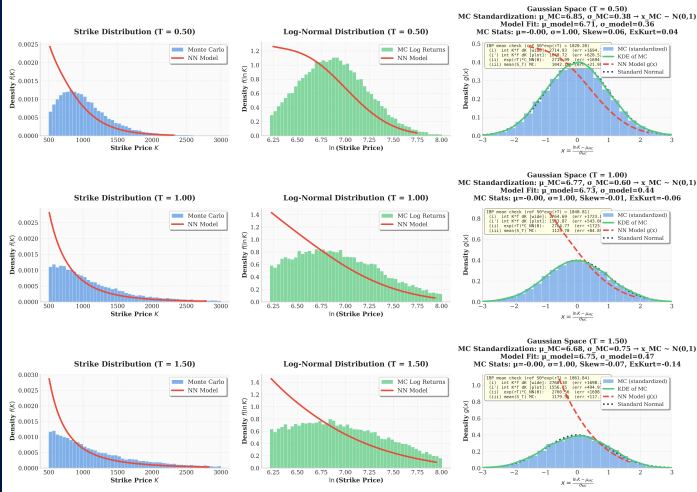
Left: legacy $\tilde{\varphi}$ (pre-correction). Centre: pretrained corrected. Right: retrained K_0 BC.

Full comparison: pre-correction

PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S_t) S_t dW_t$

Pre-correction (legacy phi_tilde mapping)

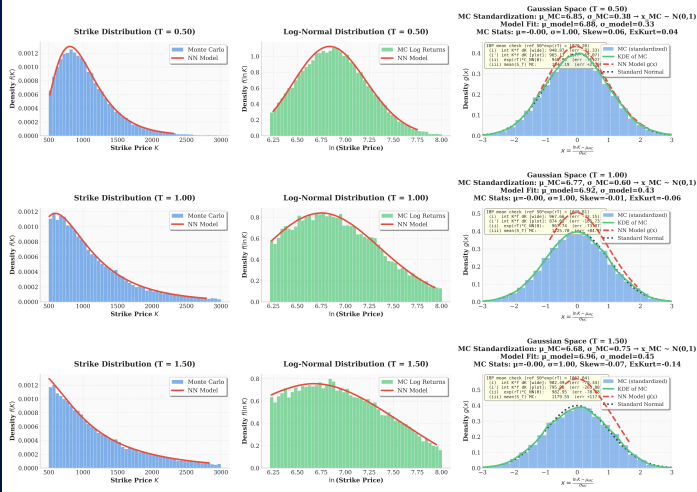


Full comparison: pretrained (corrected)

PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S_t) S_t dW_t$

Pretrained (corrected ϕ -tilde mapping)

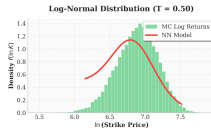


Full comparison: retrained

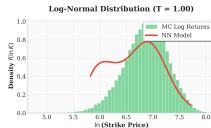
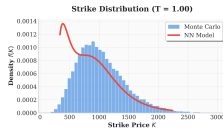
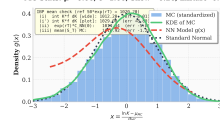
PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S_t) S_t dW_t$

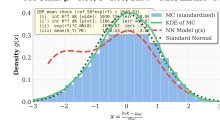
Retrained (K0 BC penalty, K_min=100)



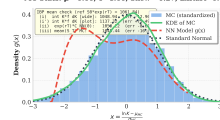
Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.88, \sigma_{MC}=0.29 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.76, \sigma_{model}=0.32$
MC Stats: $\mu=-0.00, \alpha=1.00, \text{Skew}=-0.26, \text{ExKurt}=0.11$



Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.87, \sigma_{MC}=0.41 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.69, \sigma_{model}=0.44$
MC Stats: $\mu=-0.00, \alpha=1.00, \text{Skew}=-0.29, \text{ExKurt}=0.15$



Gaussian Space (T = 1.50)
MC Standardization: $\mu_{MC}=6.84, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.64, \sigma_{model}=0.51$
MC Stats: $\mu=-0.00, \alpha=1.00, \text{Skew}=-0.27, \text{ExKurt}=0.09$



Summary

- **IBP autodiff + integration are exact:** $M_1 \equiv M_2$ to $<0.05\%$ at every T .
- **NN _{η} MC is well-calibrated:** M_3 within 0.7% of $S_0 e^{rT}$ at every T .
- **Diagnostic fix.** The earlier `mean_mc` values (1042.19 at $T = 0.5$, etc.) were the in-window MC mean: `create_enhanced_pdf_analysis` inadvertently passed the $[K_{\min}, K_{\max}]$ -trimmed sample to the IBP M_3 computation. The MC develops a heavy right tail under the local-vol surface, so asymmetric trimming biased M_3 upward by $\sim 2\%$. Patched at `dupire_pipeline.py:1590` (`mc_used` $\rightarrow$ `mc_samples`); the corrected unconditional mean is 1018.25.
- **Residual KDE-vs-NN deviation:** dominated by NN _{φ} undershoot near $K = 0$ ($M_1 \approx M_2 \approx 7\%$ below forward).
- **Fix:** $K=0$ boundary penalty in the loss (`models/runs/k0_bc_retrain`) brings all three legs within $\sim 1\%$ at $T \in \{0.5, 1.0\}$.



Questions?

Plots: [plots/ibp_comparison/](#)